# A Comparative Spectrum Selection Study of Hyperspectral and Narrowband Ultrahyperspectral Sounder for Stratospheric Temperature Remote Sensing Applications

Sufeng Wang, Feng Lu, Juan Li, Di Fu, Chenguang Chang, Bin Han, and Yutao Feng

***Abstract*—Anomalous variations in stratospheric circulation, together with stratosphere–troposphere coupling processes, exert profound modulating effects on tropospheric circulation patterns and extreme weather events. Stratospheric temperature serves as a critical indicator of the Earth system's response to climate change. Satellite-borne stratospheric temperature sounders offer extensive spatiotemporal coverage and high measurement consistency, and spectral band selection is a core issue in instrument design. This study proposes an optimal spectral band selection method for stratospheric temperature profile retrieval, and compares the sounding performance of the proposed narrow-band scheme (NBS) with that of mainstream spaceborne hyperspectral infrared sounders (IRSs). Using a sounding model constructed with a space heterodyne spectrometer (SHS), optimal narrow bands are selected within the 15 $\mu$m $CO_2$ absorption band. Accounting for the interference effects of $O_3$ and $H_2O$, the application performance of NBS in atmospheric sounding is comprehensively evaluated. Comparative analyses are conducted between NBS and typical hyperspectral IRSs, including IASI-NG, atmospheric IRS (AIRS), IRS, and geostationary interferometric IRS (GIIRS). The results demonstrate that NBS exhibits excellent sensitivity to stratospheric temperature profiles, and the errors in temperature retrieval caused by $O_3$ and $H_2O$ remain controllable. Under idealized conditions, when the noise-equivalent temperature differential (NEdT) of NBS is matched with each reference sensor, the retrieval accuracy (average RMSE) of the 0.7–200 hPa is improved by 0.30, 0.84, 2.61, and 2.54 K, and the vertical resolution is enhanced by 0.76, 2.12, 8.85, and 8.39 km, respectively.**

***Index Terms*—Atmospheric vertical sounding, space heterodyne spectrometer (SHS), spectral band selection, stratospheric temperature, temperature profile retrieval.**

Received 2 December 2025; revised 7 May 2026; accepted 26 May 2026. Date of publication 5 June 2026; date of current version 10 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42405143 and in part by West Light Cross-Disciplinary Innovation Team of Chinese Academy of Sciences under Grant xbzg-zdsys-202411. *(Corresponding authors: Feng Lu; Yutao Feng.)*

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Digital Object Identifier 10.1109/TGRS.2026.3700124

## I. Introduction

RESEARCH on stratospheric temperature has received widespread attention during the last three decades due to its close relation to climate change, ozone depletion, and greenhouse gases increase. Temperature variability in the stratosphere (~0.7–100 hPa) can influence tropospheric circulation and contribute to extreme weather events [1], [2]. It is widely recognized as a diagnostic indicator of climate forcing [3], [4], [5] as well as large-scale dynamical [6] and chemical processes [7]. In 1990, the World Meteorological Organization (WMO) conducted a comprehensive scientific assessment of stratospheric temperature variations [8]. The assessment report indicates that temperature variations have a significant impact on the physical and chemical processes occurring within the stratosphere [8]. In the mid-1990s, the Joint Scientific Committee of the World Climate Research Program (WCRP) took Stratospheric Processes and their Role in Climate (SPARC) as the core subprogram of WCRP [9] and established an assessment team for the trend of stratospheric temperature. The assessment indicates that temperature, water vapor, and ozone are indicators of climate change in the stratosphere [10], [11]. In addition, the Intergovernmental Panel on Climate Change (IPCC) attached considerable importance to trends in stratospheric temperature, taking them as a key evaluation indicator [10].

Hyperspectral infrared sounders (IRSs) onboard meteorological satellites offer a wide-coverage, high-precision remote sensing method for stratospheric temperature profile sounding. These sounders mainly include the infrared atmospheric sounding interferometer (IASI), atmospheric IRS (AIRS), IASI-new generation (IASI-NG), Geostationary Interferometric IRS (GIIRS), IRS, hyperspectral infrared atmospheric sounder (HIRAS), and cross-track IRS (CrIS) [12], [13], [14]. They are capable of retrieving multiparameter atmospheric profiles, including temperature, via broad-spectrum nadir observations. AIRS covers three spectral bands of 650–1135 $cm^{-1}$, 1215–1615 $cm^{-1}$, and 2180–2665 $cm^{-1}$, with corresponding spectral resolutions of 0.5, 1.2, and 2 $cm^{-1}$, respectively [15], [16]. IASI and IASI-NG share an identical spectral range of 645–2760 $cm^{-1}$, with spectral intervals of 0.25 and 0.125 $cm^{-1}$, respectively [17], [18], [19]. The spectral coverage of GIIRS onboard FY-4B spans 680–2250 $cm^{-1}$,

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partitioned into two spectral bands: 680–1130 $cm^{-1}$ and 1650–2250 $cm^{-1}$. Both bands adopt a uniform spectral sampling interval of 0.625 $cm^{-1}$ [20], [21]. HIRAS and CrIS both cover one band in the long-wave infrared spectrum (650–1136 $cm^{-1}$ for HIRAS and 650–1095 $cm^{-1}$ for CrIS) and two bands in the middle-wave infrared spectrum (1210–1750 $cm^{-1}$ and 2155–2550 $cm^{-1}$) at a spectral sampling interval of 0.625, 1.25, and 2.5 $cm^{-1}$, respectively [22], [23], [24]. The spectral bands of IRS are located in 680–1210 $cm^{-1}$ and 1600–2250 $cm^{-1}$, with an unapodized sampling rate of 0.6 $cm^{-1}$ [25]. Generally, the stratospheric temperature profile is measured via the 15 $\mu$m $CO_2$ absorption bands [26]. The spectral resolution of these sounders ranges from 0.125 to 0.625 $cm^{-1}$. Spectral bands sensitive to tropospheric temperature span a broad range, whereas those for stratospheric temperature are much narrower. Limited by inadequate spectral resolution, the capability of stratospheric temperature sounding is restricted. Existing studies have demonstrated that the AIRS can attain temperature retrieval accuracies of 0.5 K for the troposphere and 2–3 K for the stratosphere. [27], [28]; IASI achieves 0.5 K and 1.5–6 K [29], [30], [31]; and GIIRS achieves 0.5 K and 3–4.5 K [32]. According to the above analysis, the measurement capabilities of the current hyperspectral IRSs for stratospheric temperature are significantly inferior to those for the troposphere. This is because current operational meteorological applications, driven by practical demands, primarily focus on high-precision sounding of tropospheric atmospheric parameters. From the perspective of atmospheric spectral characteristics, hyperspectral IRSs mainly select temperature sounding bands within the 15-$\mu$m $CO_2$ absorption band (600–800 $cm^{-1}$). Within this band, the region exhibiting sensitivity to stratospheric temperature is relatively narrow, spanning only approximately 40 $cm^{-1}$. In particular, the spectral region sensitive to the upper stratosphere (0.7–10 hPa) mainly lies within 667–669 $cm^{-1}$, spanning only about 2 $cm^{-1}$ [33]. Given hardware constraints, these sounders typically achieve ~0.5 $cm^{-1}$ spectral resolution, providing insufficient channels for stratospheric temperature sounding. Consequently, they lack the ultrahigh-resolution observations required for accurate stratospheric measurements. Thus, their limited spectral resolution results in insufficient capability to measure stratospheric temperature profiles.

As demand for stratospheric temperature observations grows, the new generation of numerical weather prediction and climate change research urgently requires extending the high-precision, continuous measurements from the troposphere into the stratosphere. Our team previously reported a satellite-borne sounding method for stratospheric temperature profiles using narrow bands with ultrahigh spectral resolution in nadir observation mode [33]. This method utilizes several discontinuous narrow bands with ultrahigh spectral resolution to jointly measure temperature profiles from 0.7 to 200 hPa, covering the stratosphere primarily. Using the spaceborne space heterodyne spectrometer (SHS) as the sounder model, we adopt two narrow bands to achieve high-precision detection of stratospheric temperature profiles with a spectral resolution of 0.03 $cm^{-1}$, a spectral range of 9.6 $cm^{-1}$, and an instrument noise of 0.1 K. In 0.7–200 hPa (~11.5–50 km), the temperature retrieval accuracy (the average of RMSE) is approximately 0.9 K. The narrow band used in this measurement method directly affects the sounding capacity of stratospheric temperature profiles. Furthermore, research is still required for narrow-band observation schemes, particularly the impact of atmospheric component interference on sounding and the improvement in sounding performance comparisons with hyperspectral IRSs such as IASI-NG and AIRS.

This article puts forward a method for selecting the optimal spectral band for the temperature profile sounding in the atmospheric region of 0.7–200 hPa, encompassing the stratosphere primarily, and compares the sounding performance of the selected narrow band scheme with that of the spaceborne hyperspectral IRSs. Using satellite-borne SHS as the sounder model, the optimal narrow band is selected within the 15 $\mu$m $CO_2$ absorption band. The influence of the interfering components $O_3$ and $H_2O$ on temperature retrieval is analyzed. The sounding performance of the optimal narrow-band scheme (NBS) is compared with that of the hyperspectral IRSs IASI-NG, AIRS, IRS, and GIIRS. This article is organized in the following manner. Section II describes the radiative transfer model (RTM), data, spectral band selection method, retrieval method, and evaluation method; Section III provides interference assessment and comparison of sounding performance; Section IV presents the conclusions.

## II. RTM, Data, and Methods

### A. Radiative Transfer Model

The RTM is an indispensable computational tool for the simulations. In this article, the line-by-line RTM (LBLRTM) and radiative transfer for TOVS (RTTOV) are taken as the radiative transfer calculation tools. LBLRTM is an accurate and flexible RTM that covers the full spectral range from the microwave to the ultraviolet [34]. It has the ability to calculate optical depth, transmittance, radiance, and analytic Jacobian. It provides the foundation for many radiative transfer applications, such as the training of IASI's operational model sigma-IASI, absorption coefficient calculation for use in RTMs for GCMs for climate applications [35], and so on. RTTOV, a fast radiative-transfer package built by NWP Satellite Application Facilities [36]. It serves as an observational operator for assimilating satellite measurements into NWP models, and is also employed to train machine-learning-based approaches for simulating satellite observations [37]. RTTOV is developed based on LBLRTM. RTTOV is trained using LBLRTM calculations of line-by-line optical thickness under a wide range of atmospheric conditions. For several infrared sensors (e.g., IASI and AIRS), the infrared coefficient files of RTTOV are derived from LBLRTM line-by-line calculations [38]. For the 600–1000 $cm^{-1}$ thermal IR band, the standard deviation of brightness temperature differences between the two model calculations is <0.15 K [37].

### B. Data

The spectral band selection adopts six classic AFGL standard atmospheric profiles: the U.S. Standard Atmosphere,

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TABLE I
SEASONAL AND GEOGRAPHICAL DISTRIBUTION CHARACTERISTICS OF TEST SAMPLES

| | | Number | Proportion |
|---|---|---|---|
| Latitude | N 60 ~ 90° | 225 | 16.67 % |
| | N 30 ~ 60° | 225 | 16.67 % |
| | N 0 ~ 30° | 225 | 16.67 % |
| | S 0 ~ -30° | 225 | 16.67 % |
| | S -30 ~ -60° | 225 | 16.67 % |
| | S -60 ~ -90° | 225 | 16.67 % |
| Season | Spring | 317 | 23.48 % |
| | Summer | 378 | 28.00 % |
| | Autumn | 299 | 22.15 % |
| | Winter | 356 | 26.37 % |

midlatitude winter, subarctic winter, midlatitude summer, subarctic summer, and tropical atmosphere.

ERA5 reanalysis data provided by the European Center for Medium-Range Weather Forecast (ECMWF) are employed to analyze the variations in the volume mixing ratios of ozone and atmospheric humidity. The study covers the period from January 2022 to December 2024, with the dataset sampled at eight daily UTC synoptic times: 00:00, 03:00, 06:00, 09:00, 12:00, 15:00, 18:00, and 21:00, and a spatial resolution of $5° \times 5°$.

A 60-level atmospheric profile dataset from the ECMWF is used to assess interference effects and compare with current hyperspectral sounding instruments. To ensure uniform distribution of test samples, 10% of the dataset is randomly selected as test samples for temperature retrieval, while the remaining 90% is used as training samples to construct the a priori profile and its error covariance matrix. This approach ensures the statistical results fully reflect regional characteristics. The ensemble mean of training samples serves as the a priori basis, and the a priori error covariance matrix is derived from the statistical characteristics of training samples. Detailed distribution characteristics of test samples are presented in Table I.

### C. Spectral Band Selection Method

The spectral band selection method for stratospheric temperature sounding is based on the theories of atmospheric radiation transfer and information entropy. There are two basic criteria. One is that the selected spectral band should have sufficient information content. The other is that the number of atmospheric pressure layers that the selected spectral band can cover should be large. This method mainly consists of two parts. The first part is effective channel selection that chooses the channels with large information content. The second part is narrow band selection, which selects a narrow band that not only contains large information content but also can cover the pressure range to be measured. Fig. 1 shows the flow diagram of the method.

*1) Effective Channel Selection:* The selection of effective channels is carried out within the 15-$\mu$m $CO_2$ absorption band (the spectral range of 600–800 $cm^{-1}$). This part consists of two sections: coarse selection and fine selection. The coarse selection of channels is carried out by using atmospheric absorption and weighting functions. The fine selection of channels is executed using the information entropy iterative method.

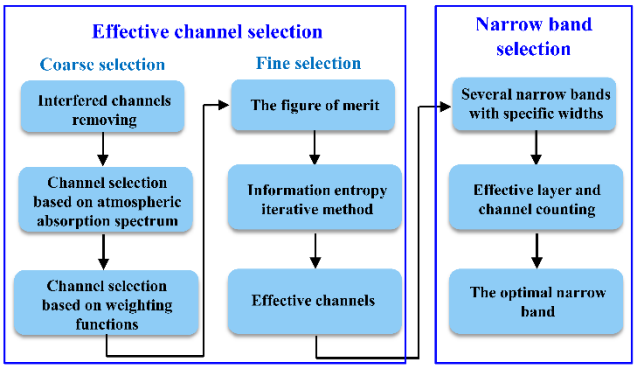



Fig. 1. Processing flowchart of the spectral band selection.

*a) Coarse Selection:*

The coarse selection is the initial screening based on the atmospheric absorption spectrum and the magnitudes and widths of the weighting function of each channel. The weighting functions refer to the partial derivative $d\tau/d\ln p$, capturing atmospheric transmittance ($\tau$) sensitivity to atmospheric pressure ($p$) [39]. For a given wavenumber, it can indicate the pressure or altitude from which the atmospheric radiation is emitted. This means the weighting functions of the channels can reflect the atmospheric information of a certain pressure, but it cannot indicate which specific atmospheric parameter (such as temperature, humidity, ozone, and so on.) is being reflected. Even so, it can be used to quickly filter out the channels that contain effective information.

The steps for coarse selection of the channel are as follows.

*Step 1:* Remove channels with significant emissions from the interference gas species, which mainly include those affected by $H_2O$ and $O_3$ within the spectral range of 600–800 $cm^{-1}$.

*Step 2:* Calculate the absorption spectrum of the atmosphere in the spectral range of 600–800 $cm^{-1}$. For those channels retained in step 1, select all the spectral channels that are located at the absorption peaks and troughs, and designate them as alternative channels (the number of these channels is denoted as $N_1$). These channels are selected because the peak channels have strong atmospheric absorption and can thus reflect the temperature in the middle and upper layers of the stratosphere. While the trough channels usually have larger peak values of the weighting function, which means they contribute more to the radiation at the corresponding pressure.

*Step 3:* For the channels remained in step 1 but not selected in the step 2, calculate the weighting function of each channel and record the effective radiative layer of the weighting function for each channel (The maximum value of the weighting function curve indicates the contribution of atmospheric radiation from a certain pressure layer to the total radiation of the channel is the greatest, and this pressure layer is referred to as the effective radiative layer [26]). Count the

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number of effective radiative layers which is denoted as $n$. For the channels that have the same effective radiative layers, sort them according to the peak value and curve width of their weighting functions. Select $m$ channels with higher peak values of the weighting function and $m$ channels with narrower weighting function curves in each effective radiative layer. Collect the selected channels of all effective radiative layers, then take them as the alternative channels (the number of channels is denoted as $N_2, N_2 \leq 2 \times m \times n$). The value of $m$ is related to the difference between $N_1$ and $N_2$. $m$ should be chosen such that $|N_2 - N_1| = 0$. Since there may be cases where no value of $m$ can make $|N_2 - N_1| = 0, m$ should make $|N_2 - N_1|$ reach the minimum.

*Step 4:* Put the channels selected by steps 2 and 3 together, and take them as the initially selected alternative effective channels (the number is denoted as $N_3$, where $N_3 = N_1 + N_2$).

*b) Fine Selection:*

The fine selection will find channels that are sensitive to temperature at specific pressures from the alternative effective channels of coarse selection. The steps are as follows.

*Step 1:* The information entropy iterative method is adopted for the selection, which is based on the information content of sounder observations. Entropy reduction (ER) is taken as the figure of merit in this study [40], [41]

$$ER = -\frac{1}{2} \ln \left| S S_a^{-1} \right| \tag{1}$$

where $S_a$ and $S$ are the a priori error covariance matrix and the retrieval error covariance matrix, respectively. $S$ can be expressed as follows:

$$S = S_a - S_a K^T \left( K S_a K^T + S_\varepsilon \right)^{-1} K S_a \tag{2}$$

where $S_\varepsilon$ denotes the measurement error covariance matrix, and $K$ denotes the temperature Jacobian.

*Step 2:* During each iteration, calculate ER of all the channels. The channel with the maximum ER is the one to be selected in this iteration. After an iteration is completed, the calculated $S$ in this iteration will be the input for the next iteration. In each iteration, one channel is selected, and this selected channel will not participate in the next iteration. When the ER of the selected channels is less than 0.01, stop the iteration. All the selected channels are the effective channels that contribute significantly to the stratospheric temperature sounding within 600–800 $cm^{-1}$ (the number of these channels is denoted as $N_4$).

*2) Narrow Band Selection:* Narrow band selection is used to select a spectral band with a certain width that can cover a specific stratospheric pressure range. The steps are as follows.

*Step 1:* Arrange the effective channels selected by section I) in ascending order of wavenumber. Starting from the channel with the smallest wavenumber, take each effective channel as the starting point, and sequentially extract narrow bands using a certain spectral width. Then multiple narrow bands can be obtained.

*Step 2:* Calculate the temperature Jacobian of the effective channels within each narrow band obtained from Step 1. Record the atmospheric pressure layer that corresponds to the peak value of the Jacobian for each channel (This pressure layer is called the "temperature-sensitive layer" in this article; it means the channel has its greatest sensitivity to the temperature in this pressure layer). Count the number of temperature-sensitive layers contained in each narrow band.

*Step 3:* Find out all the narrow bands that can cover the pressure range to be measured. Arrange these narrow bands in descending order of the number of their temperature-sensitive layers. Select the three narrow bands with the largest number of temperature-sensitive layers. Then, from these bands, choose the narrow band with the largest ER within the specified stratospheric pressure range. The finally selected narrow band is considered the optimal spectral band for the temperatures in the atmospheric pressure range to be measured.

### D. Retrieval Method

This article adopts the iterative optimal estimation method for the retrieval. This method builds on the radiative transfer equation to solve for the atmospheric parameters [42], [43]. An RTM and a priori profiles are used for calculating the measurement estimate. During the entire iteration process, the temperature state estimate is continuously adjusted to obtain the measurement estimate that approximates the actual measurement values. The final retrieval solution is obtained based on the judgment of the iterative convergence condition. This retrieval method usually requires a cost function and an optimization method. The cost function gradually decreases and stabilizes using the optimization method. When the convergence condition is met, the retrieval is completed.

This article adopts the maximum probability cost function [43], [44], as shown in the following formula:

$$\begin{aligned} J(X) = &\left(Y^m - F(X)\right)^T \times S_\varepsilon^{-1} \times \left(Y^m - F(X)\right) \\ &+ \left(X - X_a\right)^T \times S_a^{-1} \times \left(X - X_a\right) \end{aligned} \tag{3}$$

where $Y^m$ represents the brightness temperature from a sounder, $F(X)$ is the measurement estimate, $X_a$ denotes an a priori temperature profile, and $S_\varepsilon$ is the measurement error covariance, which is diagonal.

The Gauss–Newton approach is utilized as the optimization method, and the iteration format is as follows:

$$\begin{aligned} X_{n+1} = &X_a + S_a K_n^T \left( S_\varepsilon + K_n S_a K_n^T \right)^{-1} \\ &\times \left[ Y^m - F(X_n) + K_n \left( X_n - X_a \right) \right]. \end{aligned} \tag{4}$$

In this formula, $n = 0, 1, 2 \ldots$, when $n = 0, X_0 = X_a$, and when $n = 1, 2, 3 \ldots, X_n$ takes the temperature result of the previous iteration. The convergence condition is that the relative change in the cost function value is less than 1%.

### E. Evaluation Method

The sounding performance for stratospheric temperature profile is evaluated by using vertical resolution and retrieval accuracy. Vertical resolution reflects the smallest scale of atmospheric vertical structure that can be identified [45]. The data density method is adopted for this calculation. Then vertical resolution can be driven by the averaging kernel function, using the following equations [40]:

$$A = \left( K^T S_\varepsilon^{-1} K + S_a^{-1} \right) K^T S_\varepsilon^{-1} K \tag{5}$$

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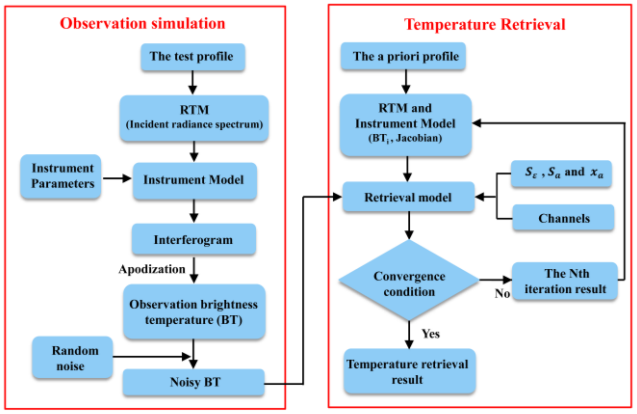



Fig. 2. Flowchart of the observation simulation and temperature retrieval.

$$R_i = \rho_i^{-1} = \delta Z_i / A_{i,i} \tag{6}$$

where $A$ represents the averaging kernel function, $R$ represents the vertical resolution, $\rho_i$ is the data density, and $\delta Z_i$ is the increment of the altitude of the $i$th layer of the atmosphere.

Two statistical metrics are employed to evaluate the retrieval accuracy, namely root-mean-square error (RMSE) and average bias (BIAS) [46]

$$RMS\,E_j = \sqrt{\frac{\sum_{i=1}^{n}\left(x_{i,j} - x'_{i,j}\right)^2}{n}} \tag{7}$$

$$BIAS_j = \frac{\sum_{i=1}^{n}\left(x_{i,j} - x'_{i,j}\right)}{n} \tag{8}$$

where $x_{i,j}$ and $x'_{i,j}$ are the $i$th retrieval temperature result and true temperature at the $j$th pressure level, respectively, and $n$ is the number of sample profiles.

## III. Results and Discussion

An SHS is a new type of instrument that can achieve ultra-high spectral resolution for radiation measurement, developed based on the Michelson interferometer [47], [48]. Compared with the structure of the Michelson interferometer, SHS replaces the flat mirror of the Michelson interferometer with a diffraction grating, thereby possessing the characteristics of spatial diffraction of the grating and the high throughput of the interferometer [49]. It provides a possibility for the highly accurate observation of the temperature profile in the stratosphere. Therefore, this article adopts the spaceborne SHS to evaluate the sounding performance of narrow band. The interferogram signal collected by SHS along the $X$-axis (diffraction direction of grating) can be written as [49]

$$I(x) = \int B(\sigma)\left(1 + \cos\left(2\pi\left(4x\left(\sigma - \sigma_0\right)\tan\theta\right)\right)\right)\mathrm{d}\sigma \tag{9}$$

where x is the location on the detector array, $B(\sigma)$ is the incident radiance spectrum, $\sigma$ is the wavenumber of the incident radiance spectrum, $\theta$ is the Littrow angle of the gratings, $\sigma_0$ and is the Littrow wavenumber. Fig. 2 shows the flowchart of the observation simulation and temperature retrieval. We use

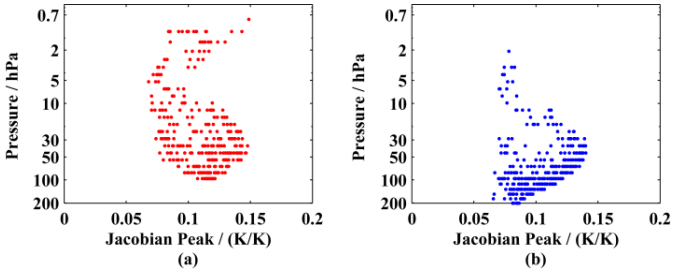



Fig. 3. Temperature Jacobians peak corresponding to atmospheric pressure. (a) Band 1. (b) Band 2.

TABLE II
System Parameters of Spatial Heterodyne Spectrometer

| Parameters | Value |
|---|---|
| Spectral region | Band 1: 666.87–676.44 $cm^{-1}$<br>Band 2: 685.05–694.62 $cm^{-1}$ |
| Spectral resolution | 0.03 $cm^{-1}$ |
| F number | 1.2 |
| Transmittance | 0.2 |
| Exposure time | 1.15 s |
| Pixel number | 640×512 |
| Pixel pitch | 30 μm |
| NEdT | ~0.1 K@ 226K |

LBLRTM to simulate the incident radiance spectrum, transform it into an interferogram using the instrument model [see (9)], apply Hamming window apodization to the interferogram, and reconstruct the radiance spectrum via Fourier transform. Transform the radiance spectrum into brightness temperature spectrum utilizing the Planck function. Gaussian-distributed random noise with a zero mean and a standard deviation equal to NEdT (noise-equivalent temperature difference) is added to the simulated observation brightness temperature [50].

Under the conditions of a certain spectral resolution, spectral range, and instrument noise, the optimal narrow band can be selected by using the method described in Section II. In our previous research, we conducted narrow band selection for spectral resolutions of 0.1, 0.05, 0.03, and 0.01 $cm^{-1}$, respectively, and proposed optimal parameter configurations. Herein, we take the optimized parameter configuration with a spectral resolution of 0.03 $cm^{-1}$ as an example to discuss the interference of other atmospheric component spectra on the selected narrow band, and compare and evaluate the temperature sounding performance of the optimal NBS with the hyperspectral sounders IASI-NG, AIRS, IRS, and GIIRS.

Table II presents the optimal parameter configuration for the long-wave infrared SHS. The spectral resolution of the instrument is set at 0.03 $cm^{-1}$. Two narrow bands in the 15 $\mu$m $CO_2$ absorption band are used for joint sounding. The spectral width of each narrow band is 9.6 $cm^{-1}$. As shown in Fig. 3, Band 1 is mainly responsible for the sounding in 0.7–80 hPa, and Band 2 is responsible for 80–200 hPa.

### A. Interference Assessment

Due to the continuity of molecular absorption, the spectral radiation produced by the interference gas species will

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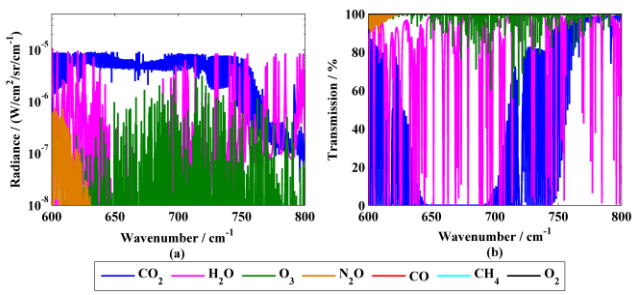



Fig. 4. Emission spectrum and transmittance of the major components of the entire atmosphere in the spectral range of 600–800 $cm^{-1}$. (a) The emission spectrum. (b) Transmittance.

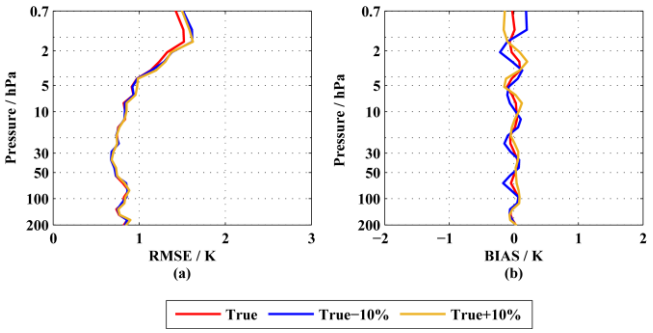



Fig. 6. Temperature retrieval results with and without changes in the volume mixing ratio of $O_3$. (a) RMSE and (b) BIAS.

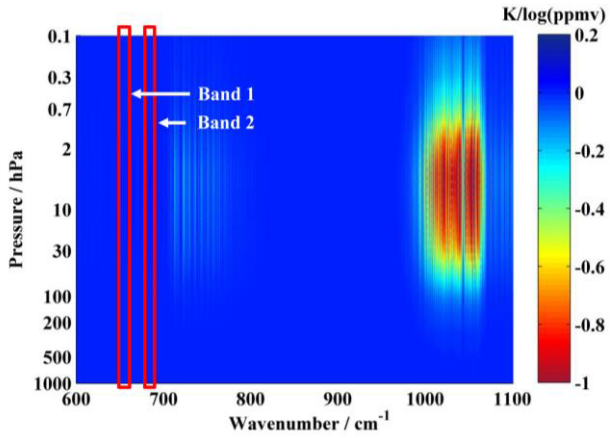



Fig. 5. Jacobians of $O_3$ in the spectral range of 600–1100 $cm^{-1}$. (Band 1 and Band 2 marked in this figure represent the sounding spectral ranges of 666.87–676.44 $cm^{-1}$ and 685.05–694.62 $cm^{-1}$).

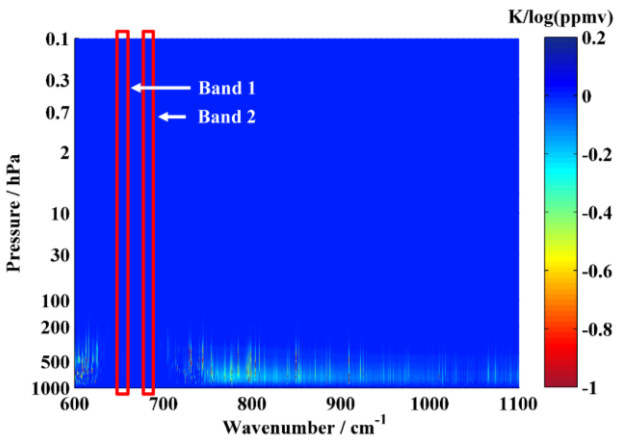



Fig. 7. Jacobians of $H_2O$ in the spectral range of 600–1100 $cm^{-1}$. (Band 1 and Band 2 marked in the figure represent the sounding spectral ranges of 666.87–676.44 $cm^{-1}$ and 685.05–694.62 $cm^{-1}$, respectively).

reduce the temperature retrieval accuracy of the instrument. Hence, it is necessary to discuss the spectral interference of other atmospheric components besides $CO_2$. Fig. 4 shows the radiation and transmittance of the main components of the entire atmosphere in the spectral range of 600–800 $cm^{-1}$. From this figure, it can be seen that there are interfering gas species in the narrow band (666.87–676.44 $cm^{-1}$ and 685.05–694.62 $cm^{-1}$), mainly $H_2O$ and $O_3$. The radiation obtained by the instrument primarily comes from $CO_2$, but $O_3$ and $H_2O$ also make certain contributions to the total radiation. Therefore, it is very important to assess the interference of $O_3$ and $H_2O$ on the temperature sounding.

*1) $O_3$:* $O_3$ is mainly distributed between 10 and 50 km. The atmosphere above 50 km becomes relatively thin and has a high transmittance, so most of the radiation produced by $O_3$ is received by the instrument. Fig. 5 shows the Jacobian matrix of $O_3$. The spectrum ranging from 1000 to 1100 $cm^{-1}$ is in the 9.6 $\mu$m $O_3$ band that has high sensitivity to $O_3$. In addition, the positions of the two narrow bands (i.e., Bands 1 and 2) are also marked in this figure. Compared with the 9.6 $\mu$m $O_3$ absorption band, the two narrow bands are almost insensitive to $O_3$.

The following explains the impact of $O_3$ on the sounding from the perspective of retrieval. This article utilizes the ERA5 data to analyze the variation in $O_3$ concentration. The results show that the average variation in the volume mixing ratio of $O_3$ is approximately 10%. The variation value is set to $\pm$10%. The retrieval results are compared for cases where the volume mixing ratio of $O_3$ remained unchanged and changed by $\pm$10%. As shown in Fig. 6, the RMSE difference between the cases with and without changes is small, within 0.1 K. The BIAS difference is slightly more pronounced below 5 hPa, with a maximum value of 0.2 K. Overall, therefore, considering 10% variations in $O_3$ concentration and the 1 K accuracy requirement for stratospheric temperature sounding in application areas specified by the WMO (https://space.oscar.wmo.int/variables), the impact of $O_3$ on NBS sounding performance (the maximum RMSE difference of 0.1 K) is controllable.

*2) $H_2O$:* $H_2O$ is primarily distributed below 10 km. Above 10 km, $CO_2$ and $O_3$ have a significant absorption effect on the radiation of $H_2O$. As shown in Fig. 4, the radiation from $H_2O$ remains relatively large at certain wavenumbers. However, Fig. 7 indicates that the Jacobian values of $H_2O$ in the two narrow bands are small. Thus, the two bands are not sensitive to $H_2O$.

The following still utilizes retrieval to illustrate the influence of $H_2O$ on the sounding. Based on the statistical results of the ERA5 data, the average change in volume mixing ratio of $H_2O$ is about 20%. The variation value is set to

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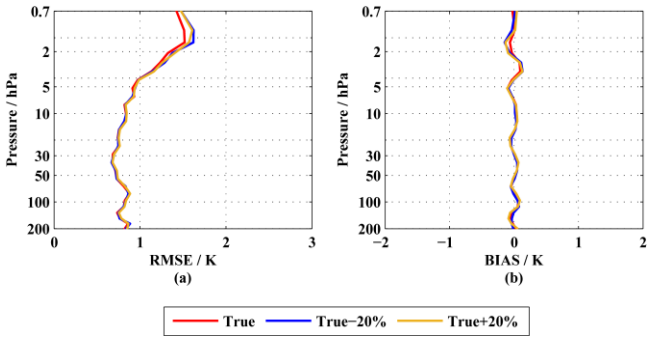



Fig. 8. Temperature retrieval results with and without changes in the volume mixing ratio of $H_2O$. (a) RMSE and (b) BIAS.

±20%. The retrieval results are compared for cases where the volume mixing ratio of $H_2O$ remained unchanged and changed by ±20%. In Fig. 8, the RMSE with and without changes in the volume mixing ratio of $H_2O$ nearly coincide at most atmospheric pressure levels, with differences within 0.1 K below 2 hPa, and BIAS is similar. Consequently, when considering 20% variations of $H_2O$ concentration and the 1-K accuracy requirement for stratospheric temperature sounding in application areas specified by the WMO, the impact of $H_2O$ on NBS sounding performance (the maximum RMSE difference of 0.1 K) remains within acceptable limits.

### *B. Comparison of Sounding Performance*

In order to further understand the temperature measurement performance of the NBS, this article will compare the vertical resolution and retrieval accuracy of the NBS with those of the hyperspectral IRSs IASI-NG, AIRS, IRS, and GIIRS. Indeed, the temperature measurement performance is affected by multiple instrument factors (such as spectral configuration, noise characteristics, instrument line shapes, and spectral response functions), among which spectral configuration is a key determinant of remote sensing performance. This section primarily analyzes sounding performance differences resulting from spectral configurations under idealized conditions. Therefore, differences in other instrument factors will not be discussed in this article, except for NEdT. We will compare the vertical resolution and retrieval accuracy between NBS and these four hyperspectral IRSs under identical NEdT.

From the calculation (3)–(6), it can be known that the vertical resolution and retrieval accuracy are not only determined by the spectral resolution and the spectral range but also by the NEdT of the instrument. In practice, the NEdT of each spectral channel is different. To simplify the comparative analysis, the NEdT of the channels for temperature sounding ("temperature channels" for short) will be averaged. According to the temperature channel selection results and the spectral interference reported in related literature [25], [51], this article will select the spectral channels with wavelengths less than 770 $cm^{-1}$ as the temperature channels. That is, the spectral region involved in the calculation is 645–770 $cm^{-1}$ for IASI-NG, 650–770 $cm^{-1}$ for AIRS, 680–770 $cm^{-1}$ for IRS, and 680–770 $cm^{-1}$ for GIIRS. Based on the NEdT data of these four sounders reported in the literature [25], [33], [52] and the Met Office 1-D-Var software (https://nwp-saf.eumetsat.int/site/software/1d-var/), the average NEdT of IASI-NG, AIRS, IRS, and GIIRS can be calculated to be approximately 0.10, 0.38, 0.50, and 0.43 K, respectively. The NEdT of the NBS is set to be equal to that of each of these four sounders, respectively.

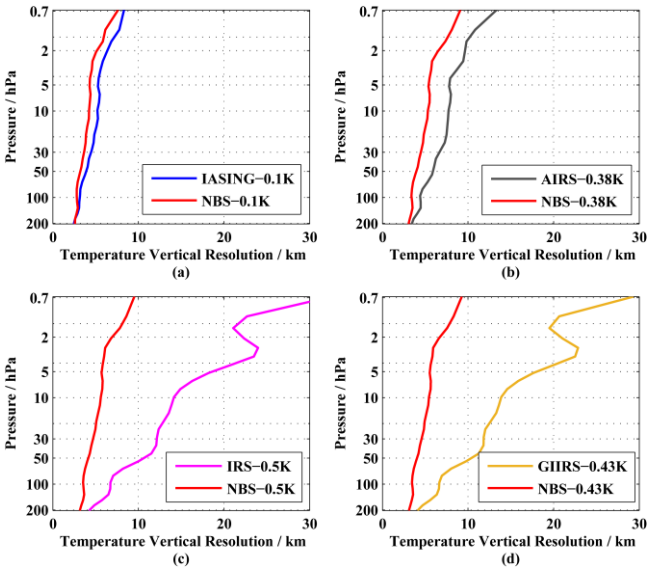



Fig. 9. Comparison results of temperature vertical resolution. (a) IASI-NG and NBS (NEdT = 0.1 K). (b) AIRS and NBS (NEdT = 0.38 K). (c) IRS and NBS (NEdT = 0.5 K). (d) GIIRS and NBS (NEdT = 0.43 K).

TABLE III
COMPARISON OF VERTICAL RESOLUTION

| Sounder | NEdT / K | Spectral band of temperature / $cm^{-1}$ | Spectral resolution / $cm^{-1}$ | Average vertical resolution / km |
|---|---|---|---|---|
| IASI-NG | 0.1 | 645–770 | 0.125 | 4.74 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 3.98 |
| AIRS | 0.38 | 650–770 | 0.5 | 7.06 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 4.94 |
| IRS | 0.5 | 680–770 | 0.6 | 14.05 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 5.20 |
| GIIRS | 0.43 | 680–770 | 0.625 | 13.45 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 5.06 |

For IASI-NG, GIIRS, AIRS, and IRS, we use RTTOV, which incorporates the spectral response functions and instrument line shapes, to directly simulate their observed brightness temperatures and Jacobians. Since the simulations are noise free, we add Gaussian-distributed random noise with a zero mean and a standard deviation equal to NEdT for these four sounders.

*1) Vertical Resolution:* Fig. 9 and Table III present the results of vertical resolution. As illustrated, when the NEdT is 0.1 K, the mean vertical resolutions of IASI-NG and the NBS in 0.7–200 hPa are 4.74 and 3.98 km, respectively. When the NEdT is 0.38 K, the corresponding values of AIRS

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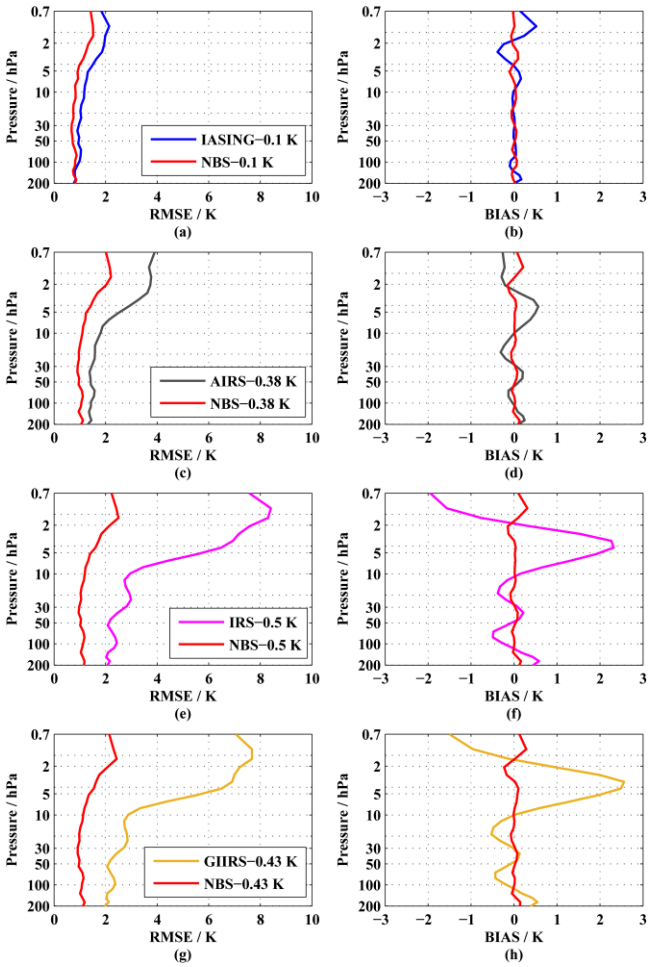



Fig. 10. Comparison results of the temperature retrieval. (a) IASI-NG and NBS (NEdT = 0.1 K). (b) AIRS and NBS (NEdT = 0.38 K). (c) IRS and NBS (NEdT = 0.5 K). (d) GIIRS and NBS (NEdT = 0.43 K).

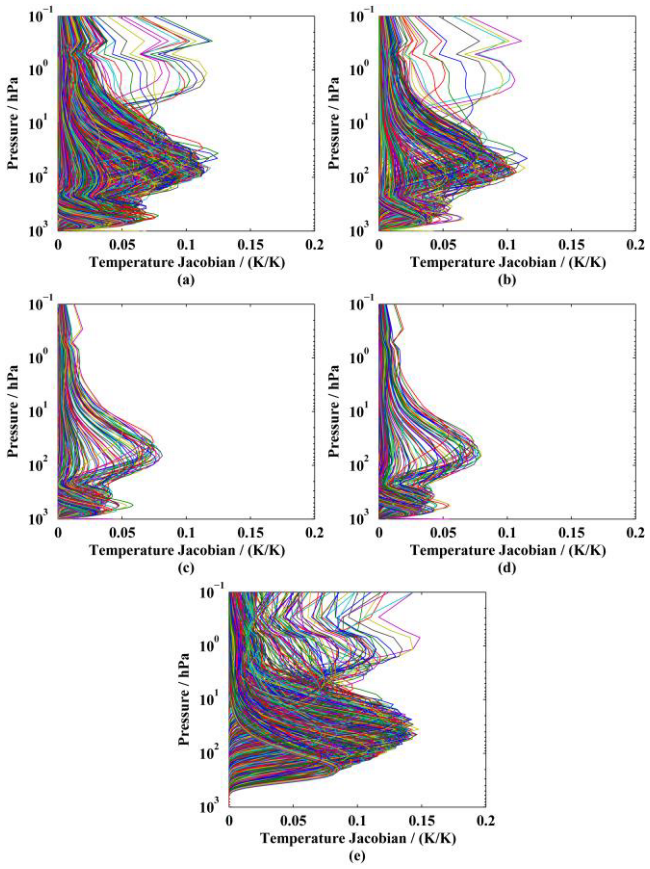



Fig. 11. Temperature Jacobian curves of the channels. (a)–(e) represent IASI-NG, AIRS, IRS, GIIRS, and NBS, respectively.

TABLE IV
COMPARISON OF RETRIEVAL ACCURACY

| Sounder | NEdT / K | Spectral band of temperature / $cm^{-1}$ | Spectral resolution / $cm^{-1}$ | Average RMSE / K |
|---|---|---|---|---|
| IASI-NG | 0.1 | 640–770 | 0.125 | 1.23 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 0.93 |
| AIRS | 0.38 | 650–770 | 0.5 | 2.10 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 1.26 |
| IRS | 0.5 | 680–770 | 0.6 | 3.98 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 1.37 |
| GIIRS | 0.43 | 680–770 | 0.625 | 3.85 |
| NBS | | 666.87–676.44<br>685.05–694.62 | 0.03 | 1.31 |

and the NBS are 7.06 and 4.94 km, respectively. When the NEdT is 0.5 K, the values of IRS and the NBS are 14.05 and 5.20 km, respectively. Finally, when the NEdT is 0.43 K, the values of GIIRS and the NBS are 13.45 and 5.06 km, respectively. Compared with IASI-NG, AIRS, IRS, and GIIRS, the NBS shows average vertical resolution improvements of 0.76, 2.12, 8.85, and 8.39 km, respectively, under idealized conditions. Thus, the temperature vertical resolution of the NBS is superior to that of IASI-NG, AIRS, IRS, and GIIRS.

*2) Retrieval Accuracy:* The retrieval results are shown in Fig. 10 and Table IV. The RMSE results indicate that, under the condition of the same NEdT, the RMSE of the NBS is superior to that of the four sounders at most pressures. In 100–200 hPa, the difference between the NBS and the four sounders in RMSE is small, particularly for IASI-NG and AIRS. In 0.7–100 hPa, the difference in RMSE increases as the pressure decreases, reaching the maximum in 2 hPa. Compared with IASI-NG, AIRS, IRS, and GIIRS, the average RMSE of the NBS in 0.7–200 hPa is improved by 0.30, 0.84, 2.61, and 2.54 K, respectively, under idealized conditions. The BIAS results show that their BIAS values are distributed on both sides of 0 K, but the NBS exhibits a smaller deviation from 0 K compared to the four sounders. Fig. 11 presents the temperature Jacobian for the temperature channels within their spectral bands. In 100–1000 hPa, IASI-NG, AIRS, IRS, and GIIRS have many sensitive channels, but the NBS has fewer sensitive channels (especially in the troposphere, where there are almost no sensitive channels). Therefore, the retrieval accuracy of the NBS is comparable to or even slightly worse

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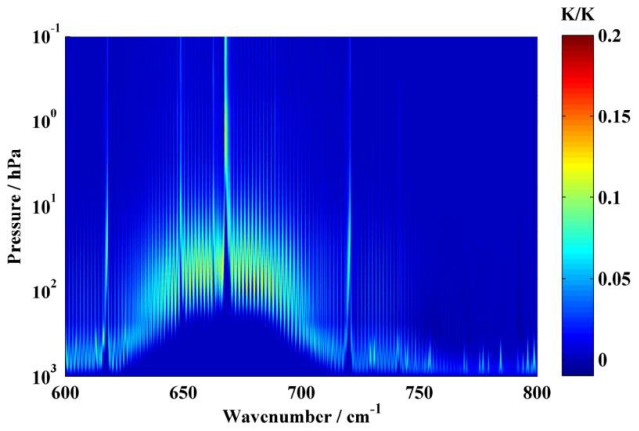



Fig. 12. Temperature Jacobians of the spectrum of 600–800 $cm^{-1}$.

than that of IASI-NG, and superior to AIRS, IRS, and GIIRS in 100–200 hPa. In 0.7–100 hPa, the sensitive channels of the NBS are abundant; IASI-NG and AIRS also have a large number of sensitive channels, which are mainly concentrated between 10 and 100 hPa, but the channels sensitive to the temperatures of 0.7–10 hPa are still relatively few; the channels of IRS and GIIRS are mainly sensitive to the pressure range of 10–100 hPa, but their sensitivity to the temperatures of 0.7–10 hPa is significantly inferior to that of IASI-NG, AIRS, and NBS. Besides, the temperature Jacobian values of some NBS channels are larger than those of the four, indicating a higher sensitivity to the temperature in the corresponding pressures. Therefore, in 0.7–100 hPa, the retrieval accuracy of NBS is significantly better than that of the four.

As shown in Fig. 12, in the long-wave infrared region ranging from 600 to 800 $cm^{-1}$, the spectral region sensitive to the stratospheric temperature is primarily concentrated between 645 and 690 $cm^{-1}$. In particular, the spectral region sensitive to the upper stratosphere (0.7–10 hPa) is predominantly located between 667 and 669 $cm^{-1}$, with a spectral width of 2 $cm^{-1}$ approximately. In terms of sounding spectral band setting, IRS and GIIRS start from 680 $cm^{-1}$ with a spectral resolution of 0.6 and 0.625 $cm^{-1}$. This configuration only covers a small portion of the spectrum sensitive to stratospheric temperature. As a result, both of them exhibit a lack of channels in the stratosphere, especially in the pressures from 0.7 to 10 hPa. The spectral bands of IASI-NG and AIRS begin with 645 and 650 $cm^{-1}$, with the spectral resolutions of 0.125 and 0.5 $cm^{-1}$, respectively. They can cover a large portion of the sensitive spectrum in the stratosphere. However, due to the limitation of their spectral resolutions and the narrow range of the spectrum sensitive to 0.7–10 hPa, they still show insufficient sensitive channels in this pressure range. NBS has two narrow bands (666.87–676.44 $cm^{-1}$ and 685.05–6 94.62 $cm^{-1}$). The spectral resolution is 0.03 $cm^{-1}$. The narrow bands are sensitive to temperatures of different pressure layers. The narrow band ranging from 666.87 to 676.44 $cm^{-1}$ is sensitive to temperatures in 0.7–80 hPa, while the other band, spanning from 685.05 to 694.62 $cm^{-1}$, is sensitive to t temperatures in 80–200 hPa. Moreover, the ultrahigh spectral resolution provides abundant channels for the temperature sounding of the whole stratosphere, thereby improving the retrieval performance [33], [53]. Therefore, NBS has a superior sounding performance for stratospheric temperature compared to IASI-NG, AIRS, IRS, and GIIRS.

## IV. Conclusion

This article presents a method for selecting optimal spectral bands for stratospheric temperature sounding and compares the performance of the proposed NBS with that of current spaceborne hyperspectral IRSs. Based on a sounder model constructed using SHS, two optimal narrow bands (666.87–676.44 $cm^{-1}$ and 685.05–694.62 $cm^{-1}$) are identified within the 15 $\mu$m $CO_2$ absorption band and exhibit high sensitivity to stratospheric temperature. The influences of $O_3$ and $H_2O$ on the temperature sounding of NBS are systematically investigated. Results from Jacobian matrix calculations and temperature retrieval experiments demonstrate that the two selected narrow bands are nearly insensitive to variations in the volume mixing ratio of $O_3$ and $H_2O$. Considering the concentration fluctuations of these gases and the 1-K accuracy requirement for stratospheric temperature sounding specified by the WMO, the impact of $O_3$ and $H_2O$ on retrieval accuracy is controllable, thereby ensuring the reliability of the NBS in temperature measurement. The sounding performance of the NBS is compared with that of four mainstream spaceborne hyperspectral sounders, namely IASI-NG, AIRS, IRS, and GIIRS. In practical applications, the performance of temperature sounding is affected by multiple instrument-related factors, among which spectral configuration is a key determinant of remote sensing performance. This study focuses on analyzing the differences in sounding performance induced by different spectral configurations under idealized experimental conditions (i.e., excluding external interference factors). The results indicate that when the NEdT of NBS is consistent with that of each of the four hyperspectral sounders, the NBS outperforms them in both vertical resolution and retrieval accuracy. Under idealized conditions, within the 0.7–200 hPa pressure range (corresponding to the stratospheric region), the mean RMSE of the NBS is significantly lower than that of the four hyperspectral sounders, with respective improvement amplitudes of 0.30, 0.84, 2.61, and 2.54 K. Meanwhile, the vertical resolution of the NBS is improved by 0.76, 2.12, 8.85, and 8.39 km compared to the four sounders.

The proposed spectral band selection method can effectively identify bands highly sensitive to stratospheric temperature within the target pressure range, thereby improving the effective utilization of the sounding spectrum and enhancing the measurement capability of the sounder. This study provides important theoretical support and technical preparation for extending the observation range of satellite-borne high-precision hyperspectral remote sensing systems from the troposphere to the stratosphere, laying a solid foundation for subsequent research on stratospheric temperature remote sensing and related application fields.

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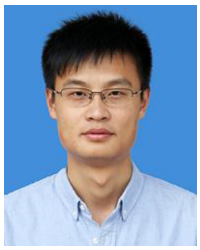
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